OncoContour

Geospatial Cancer Analytics Platform

User Guide

Version 1.1

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Genomics and Machine Intelligence Lab

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1. Introduction

OncoContour is a web-based geospatial cancer analytics platform that enables researchers, clinicians, and public health professionals to visualize, explore, and analyse cancer incidence data across geographic regions. By combining interactive heatmaps, demographic breakdowns, and multi-year trend analysis, OncoContour transforms raw cancer statistics into actionable, map-based insights.

1.1 Key Capabilities

- Interactive geographic heatmaps of population density and cancer incidence
- Per-city cancer case markers with trend charts and type breakdowns
- Comparative multi-city Plotly trend charts across user-defined year ranges
- Demographic analysis: race/ethnicity, age group, and sex distribution pie charts
- Custom CSV data import for any US geographic region
- Support for both State-based geocoding (via census data) and direct coordinate input

1.2 System Requirements

- Modern web browser (Chrome, Firefox, Edge, or Safari)
- Network access to the server running the Flask application
- No additional software installation required for end users

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2. Application Navigation

OncoContour has four main pages accessible from the navigation bar at the top of every page.

Page	Description
Home	Landing page with the OncoContour logo, application description, and quick-access previews of the two built-in Rhode Island cancer maps. Click any preview to open the full interactive map.
Import Data	Upload your own CSV datasets (cancer statistics, city coordinates, county race/ethnicity, and age/sex population data) and generate a custom visualisation report.
User Guide	Opens this document (PDF).
About	Project overview, authors, contact information, and lab website link.

3. Home Page

The Home page is the entry point to OncoContour. It displays:

- The OncoContour logo and a short application description.
- Two side-by-side iframe previews of the built-in Rhode Island cancer maps (Population Distribution and Cancer Incidence). Click either preview to navigate to the full interactive map.
- A dynamically updated footer showing the current year.

> *The built-in Rhode Island maps are pre-generated and load immediately without requiring any data upload.*

4. Importing Your Data

The Import Data page allows you to upload up to four CSV files and then generate a fully interactive visualisation report. Each file type is independent — you may upload any combination.

4.1 Cancer Statistics Data (State-based)

Use this upload when your data includes US State abbreviations but not explicit coordinates. Please note that this can exclusively map cities in a census designated place. OncoContour will geocode these cities automatically using the bundled census dataset.

Required column order

Column	Description
`City`	`City name (must match census data)`
`State`	`2-letter US state abbreviation (e.g. RI, MA, NY)`
`CancerType1 … CancerTypeN`	`One column per cancer type, integer case counts`
`Year1 … YearN`	`4-digit year columns, integer total cases for that year`

Example

```
City,State,Bladder,Breast,Lung,Prostate,2015,2016,2017,2018,2019,2020,2021
Providence,RI,11,50,40,30,47,45,44,42,41,39,37
Lincoln,RI,32,20,15,10,18,35,16,15,14,13,12
Barrington,RI,33,25,18,12,23,55,21,20,19,18,17
```

> *Cancer type columns must contain only alphabetic characters. Year columns must be 4-digit integers. A census lookup file (processed_census_data.csv) must exist in the application root for geocoding to succeed.*

4.2 City Coordinates Cancer Data

Use this format when you want to include towns that are not census designated places. You must provide latitude/longitude coordinates and population figures directly. No external census file is required.

Required column order

Column	Description
`City`	City name (display label)
`Population`	Integer population count
`Latitude`	Decimal latitude (e.g. 41.8236)
`Longitude`	Decimal longitude (e.g. -71.4222)
`CancerType1 … CancerTypeN`	One column per cancer type, integer case counts
`Year1 … YearN`	4-digit year columns, integer total cases for that year

Example

```
City,Population,Latitude,Longitude,Bladder,Breast,Lung,Prostate,2015,2016,2017,2018,2019,2020,2021
Providence,190792,41.8236,-71.4222,11,50,40,30,47,45,44,42,41,39,37
Barrington,17061,41.7409,-71.3084,33,25,18,12,23,55,21,20,19,18,17
Cranston,82635,41.7798,-71.4373,20,45,35,25,40,38,37,36,35,34,33
```

> *If both a Cancer Statistics file and a City Coordinates file are uploaded, the City Coordinates data takes priority for map generation and trend charts.*

4.3 County Race / Ethnicity Data

This file drives the race proportion pie chart and the Hispanic/Non-Hispanic breakdown chart in the report.

Required column order

Column	Description
`County`	County name
`White, Black, Asian …`	One column per race/ethnicity group; integer population counts
`Hispanic or Latino`	Optional. If present, used to build the Hispanic/Non-Hispanic pie chart automatically

Example

```
County,White,Hispanic or Latino,Black,American Indian,Asian,Pacific
Islander,Other,Two or More
Bristol,45355,1943,773,105,1285,1,500,2774
Kent,147106,9665,3220,532,4882,40,3860,10723
Newport,72063,5592,2840,362,1570,64,2533,6211
Providence,402194,160323,53803,5362,28614,380,94730,75658
Washington,116202,4578,1532,1024,2610,51,1992,6428
```

4.4 Age and Sex Population Data

This file drives the age distribution and sex proportion pie charts in the report.

Required column order

Column	Description
`Sex`	`Row identifier: must be exactly 'Male' or 'Female'`
`0-9, 10-19 … 80+`	`One column per age band; integer population counts for that sex`

Example

```
Sex,0-9,10-19,20-29,30-39,40-49,50-59,60-69,70-79,80+
Male,55897,68369,78277,74753,63778,73199,67452,38527,16969
Female,54344,66057,75425,72340,62843,75137,73676,46375,30832
```

4.5 Upload Workflow

- Click the file picker next to the relevant data section.
- Select your CSV file.
- Click the corresponding Upload button. A green confirmation or red error message appears immediately.
- Repeat for any additional files.
- Once at least one file has been successfully uploaded, click Generate Visualization.
- OncoContour validates, processes, and redirects you to the Visualization Report page.

> *Each file type is stored under a fixed name (e.g. cancer_data.csv). Re-uploading a file of the same type overwrites the previous one. All uploads are cleared when the server restarts.*

5. Visualization Report

After clicking Generate Visualization, OncoContour produces a self-contained HTML report. The report is divided into the following sections, each appearing only when the relevant data has been uploaded.

5.1 Interactive Maps

For each uploaded cancer dataset (City Coordinates or State-based), two Folium maps are generated side-by-side:

Map	What it shows
Population Distribution	Log-normalised heatmap of population density across uploaded cities. No city markers are shown — the heatmap alone conveys relative density.
Cancer Incidence Heatmap	Heatmap intensity = cases per 100,000 residents (normalised). Each city also has a clickable blue marker. Click a marker to open a popup showing the cancer-type breakdown table and a per-city trend chart.

> *Zoom and pan work normally inside each embedded map. The layer control (top-right of each map) lets you toggle the heatmap layer on and off.*

5.2 Cancer Data Table

A formatted HTML table showing the raw values from your uploaded cancer CSV (cancer type columns and year columns). Population, Latitude, and Longitude columns are hidden for clarity.

5.3 Interactive Comparative Cancer Trends

A Plotly line chart plotting total cancer cases per year for every city in your dataset on a single interactive graph. Hover over any point to see the city name, year, and case count. Use the legend to show or hide individual cities. The toolbar (top-right) provides zoom, pan, and download options.

5.4 Individual Cancer Trends by City

A grid of small Matplotlib line charts — two per row — each showing the year-over-year total cancer case trend for one city. These charts are rendered as embedded PNG images.

5.5 Race Population by County

An HTML table listing the race/ethnicity population counts per county, followed by two pie charts:

- Race Proportions — each slice is a race/ethnicity category, sized by its share of the total population across all counties.
- Hispanic or Latino vs. Non-Hispanic or Latino — if a 'Hispanic or Latino' column is present in your data, real figures are used; otherwise, Rhode Island state-level estimates are applied as a fallback.

5.6 Population Data by Age and Sex

An HTML table of age-band counts by sex, followed by two pie charts:

- Age Distribution — total population (male + female combined) broken down by age band.
- Population by Sex — total male vs. total female population.

6. Built-in Rhode Island Maps

Two pre-generated interactive maps of Rhode Island are available directly from the Home page without any data upload:

- rhode_island_cancer_map_v12.1.html — Population distribution heatmap.
- rhode_island_cancer_map_v12.html — Cancer incidence heatmap with city markers.

Click either iframe preview on the Home page to open the full-screen interactive version. These navigate to static HTML files and do not depend on uploaded data. The intention is to demonstrate (using mock data for Rhode Island) the graphs and maps that the software provides for user imported data.

7. Data Validation Rules

OncoContour validates every uploaded file before saving it. The following rules apply:

7.1 Cancer Statistics (State-based)

- Must have at least three columns: City, State, and one data column.
- First column header must be City (case-insensitive).
- Second column header must be State (case-insensitive).
- All State values must be valid 2-letter US state abbreviations (e.g. RI, MA).
- All subsequent column names must be either purely alphabetic (cancer type) or purely numeric (year).

7.2 City Coordinates Cancer Data

- First four columns must be exactly: City, Population, Latitude, Longitude.
- At least one additional column (cancer type or year) must be present.

7.3 County Race / Ethnicity Data

- First column header must be exactly County.

7.4 Age and Sex Data

- First column header must be exactly Sex.
- Sex column values should be Male and Female for pie chart generation to work correctly.

> *If a file fails validation, an error message is displayed in the upload panel. The file is not saved and any previous file of the same type is preserved.*

8. Tips & Troubleshooting

8.1 Cities not appearing on the map

- Ensure the City and State values exactly match entries in processed_census_data.csv.
- Consider switching to the City Coordinates format (Section 4.2) to bypass geocoding.

8.2 Trend charts are blank

- Year columns must be 4-digit integers (e.g. 2019, not '19' or 'Year 2019').
- Verify there are no blank or non-numeric cells in the year columns.

8.3 Hispanic pie chart uses default values

- Add a column named 'Hispanic or Latino' (exact spelling) to your County Race CSV.
- Without this column, the chart falls back to Rhode Island state-level figures (16.6% / 83.4%).

8.4 Re-uploading data

- Uploading a new file of the same type immediately replaces the previous one.
- Click Generate Visualization again to rebuild the report with the new data.

8.5 Browser caching

- OncoContour sets Cache-Control: no-store on all Flask-served pages.
- If you see a stale report, perform a hard refresh (Ctrl+Shift+R / Cmd+Shift+R).

8.6 Supported file format

- Only .csv files are accepted. Excel files (.xlsx) must be exported as CSV before uploading.
- Ensure column headers do not contain leading or trailing spaces.

9. Contact & Support

For questions, bug reports, or feature requests, please reach out to the development team:

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